Smart Home Energy Dashboard

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Problem Overview / System Description

- This system models a dashboard for the function of a smart home.
- The system is important to engineering as it holds data of energy consumption for a household. Models like this provide as a data storage as well as show the data in effect and how much change is happening in the data.
- As such, knowledge of their operations is a great and essential skill.
- Examples of such systems include:
  - Power plant controls
  - AI model rendering software
  - Larger security systems

Engineering Objective

- This system analyzes the energy usage and efficiency of a smart home.
- This ensures peak energy efficiency and optimized operational conditions.
- Examples from the graphs include:
  - Energy consumption monitoring
  - Impact on cost savings

Mathematical Model

- Our governing equation from the code is Usage = 0.5 + 3*rand(size(days)) + (hours > 17 & hours < 21)*2
- 0.5 is the baseline constant of kWh
- 3*rand() acts like the small variations in power usage in each time period
- The (hours) part accounts for higher usage at night.

Algorithm / System Logic

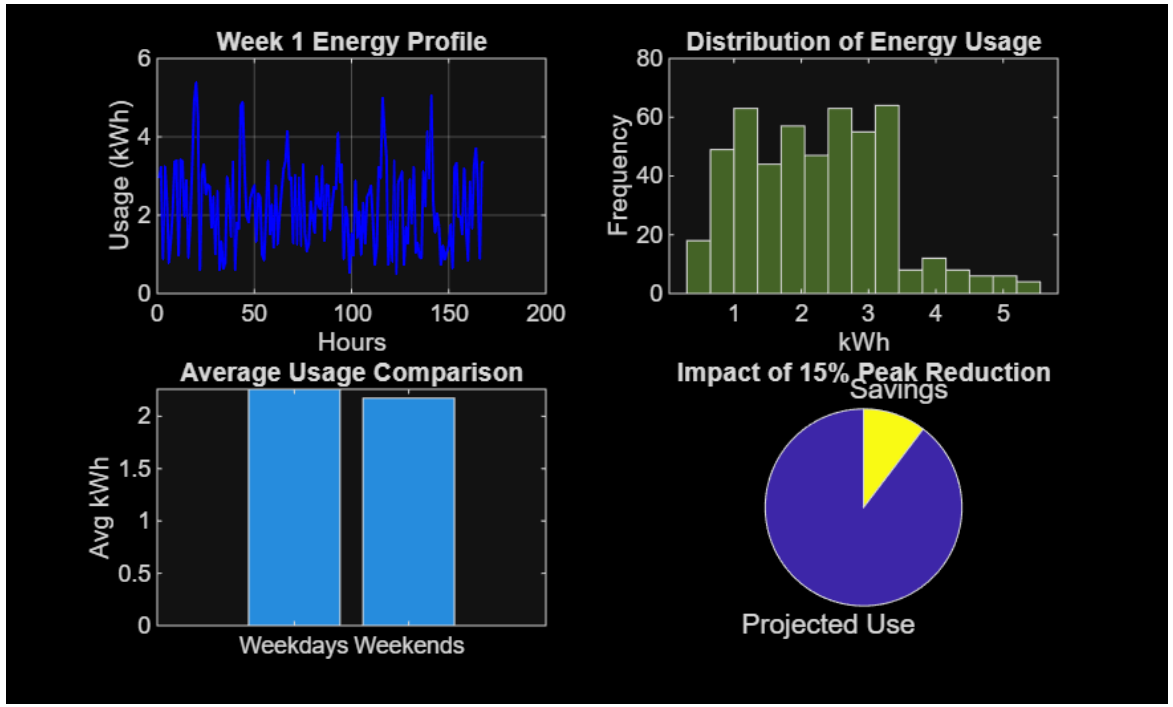
- The system works by first defining the .csv data imported into weekdays and weekends.
- The data is processed and the maximum consumption times are found, alongside the minimums.
- From there, the data is computed into a readable graph for energy consumption.

MATLAB Implementation

```
if ~isfile(filename)
days = reshape(repmat(1:21, 24, 1), [], 1);
hours = repmat((0:23)', 21, 1);
usage = 0.5 + 3*rand(size(days)) + (hours > 17 & hours < 21)*2;
T = table(days, hours, usage, 'VariableNames', {'Day', 'Hour', 'Usage_kWh'});
writetable(T, filename);
fprintf('Generated synthetic dataset: %s\n', filename);
end
data = readtable(filename);
weekday_data = data.Usage_kWh(data.IsWeekend == 0);
weekend_data = data.Usage_kWh(data.IsWeekend == 1);
avg_weekday = mean(weekday_data);
avg_weekend = mean(weekend_data);
```

Analysis of Results

- The graph logs the usage of energy of a smart home in an understandable manner.
- This is to always ensure optimal efficiency and savings, even when the temperature is consistently variable.



Model Parameters And Limitations

- Nothing major changes when parameters are adjusted.
- The graphs simply use different data or log different results according to necessary energy consumption or monetary savings.
- We assumed that the data input would work off a completed .csv file.
- This does not include incomplete or erroneous logs in temperature calculations.
- As such, a system like this requires maintenance in sensors and inaccuracies can be detrimental to savings and efficiency.

Engineering Relevance

- Systems like this are common in the engineering and computer science fields.
- As such, many important applications can be used from programs like this.
- Examples include:
  - Budgeting programs for federal departments
  - Analysis of data related to climate change in a visualization

Conclusion

- We built a system that accurately monitors the energy used by a smart home continually.
- We learned how to code functions in MATLAB and to display information taken in a graphical format.
- Knowledge of displaying data graphically is an essential skill in the engineering field and this was a great introduction.